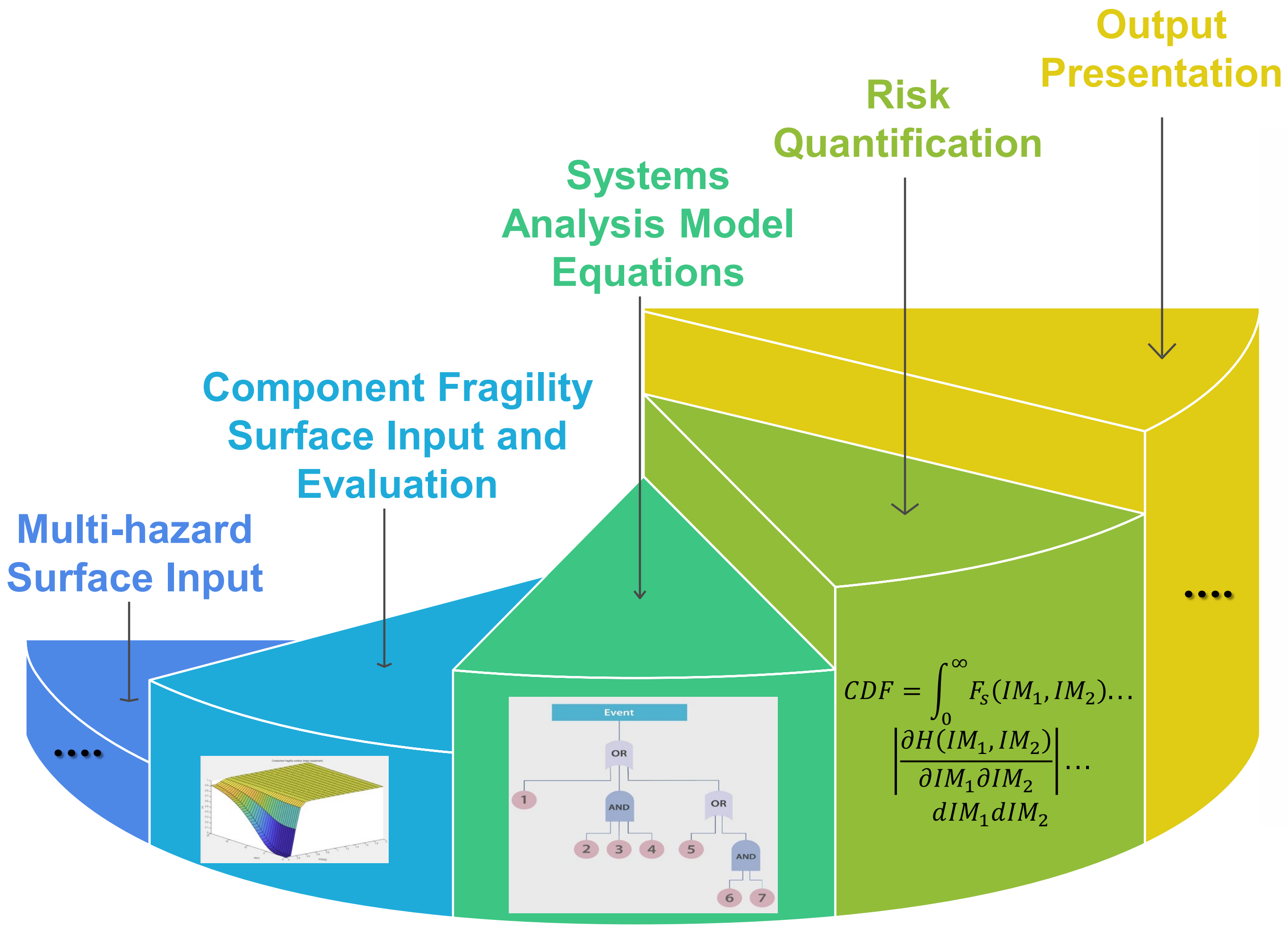
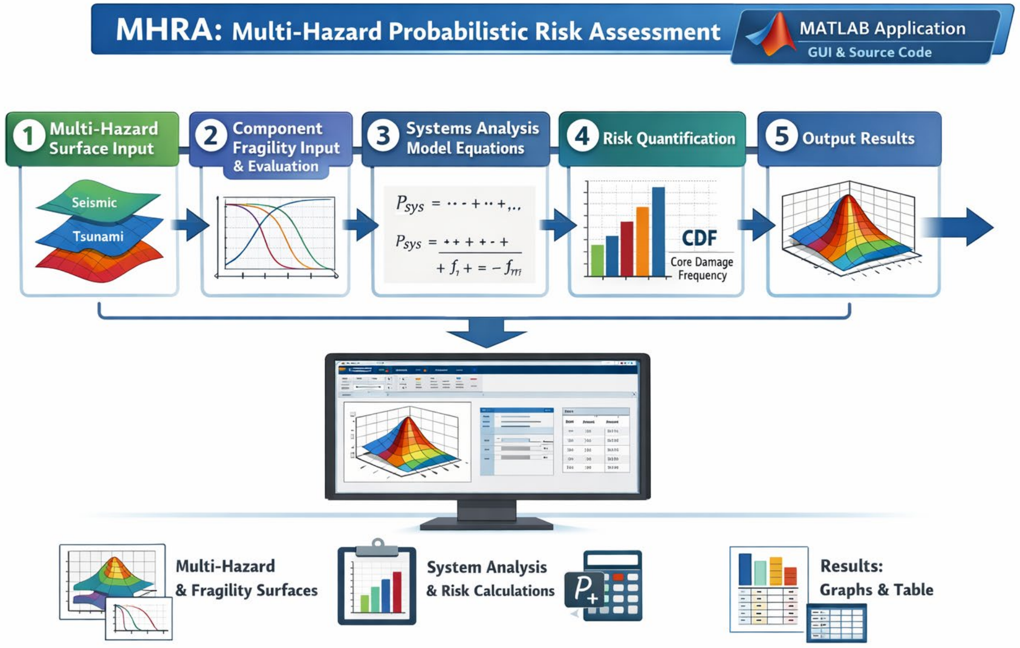
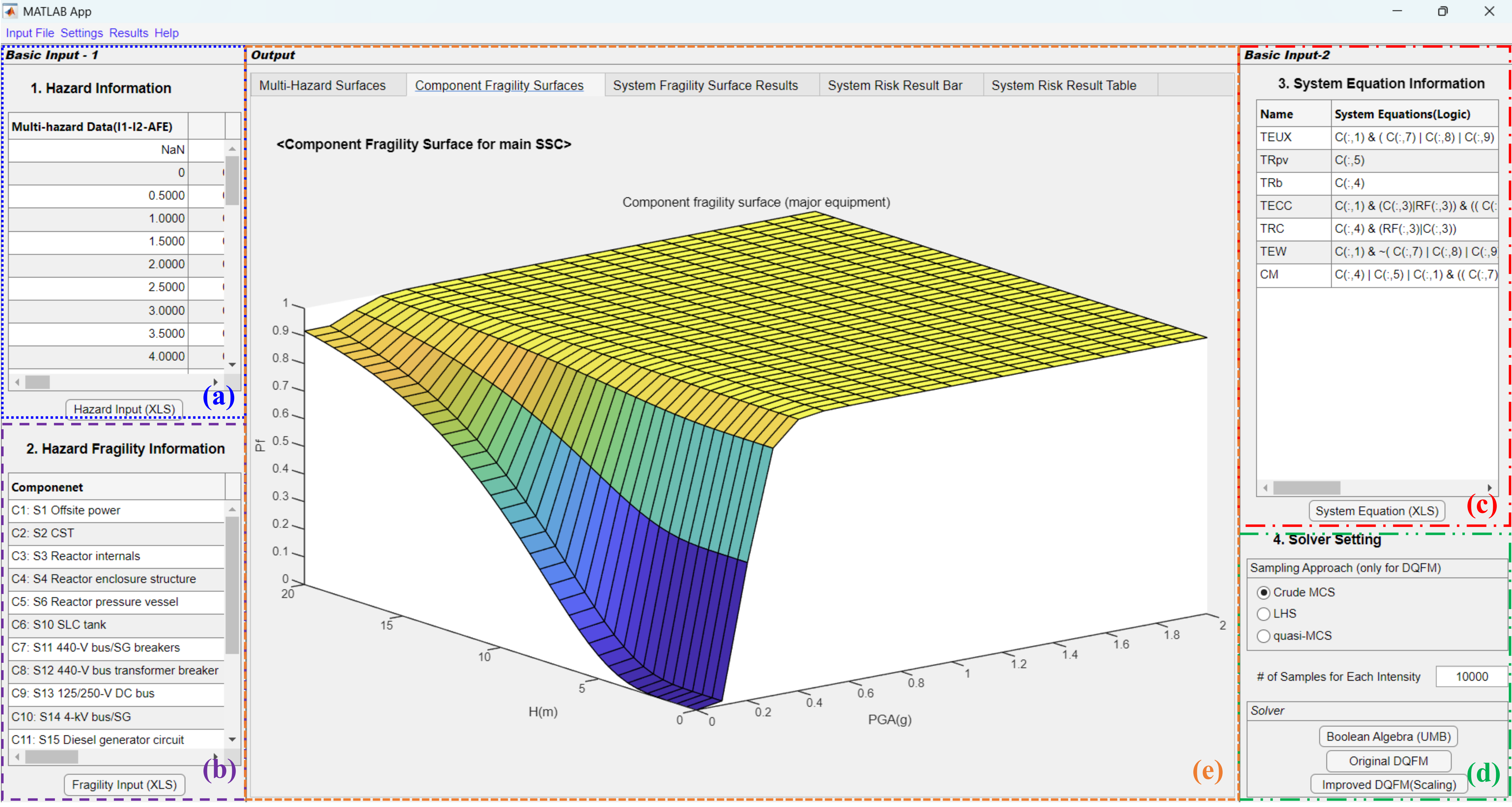
MHRA Structure

MHRA Program Modules





Matlab GUI Application Configuration



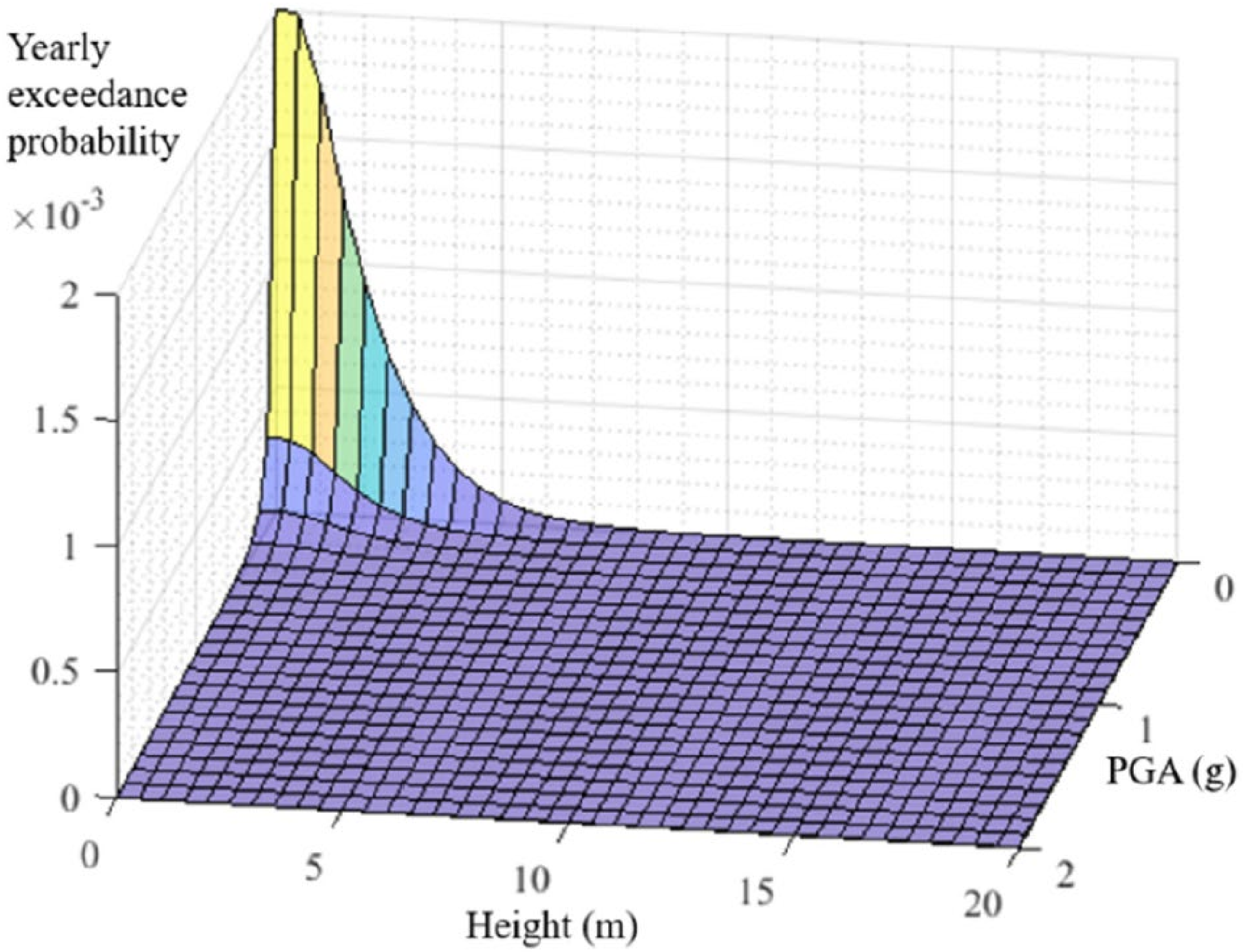


Fig. 6. Earthquake–tsunami hazard information for the numerical example (adopted from Refs. [19,23]).

Kwag, S., Ha, J. G., Kim, M. K., & Kim, J. H. (2019). Development of efficient external multi-hazard risk quantification methodology for nuclear facilities. Energies, 12(20), 3925.
Choi, E., Kwag, S., Ha, J. G., & Hahm, D. (2021). Development of a two-stage DQFM to improve efficiency of single-and multi-hazard risk quantification for nuclear facilities. Energies, 14(4), 1017.
Choi, E., Kwag, S., & Hahm, D. (2024). Multihazard capacity optimization of an NPP using a multi-objective genetic algorithm and sampling-based PSA. Nuclear Engineering and Technology, 56(2), 644-654.
Choi, E., Kwag, S., Kim, J. H., Ha, J. G., Hahm, D., & Kim, M. (2024). A review of COHRISK: Multihazard risk quantification software for nuclear power plants. Nuclear Engineering and Technology, 56(12), 5281-5290.

Table 1

Seismic fragility and random failure probability information of LGS NPP components (adapted from Ref. [19]).

Component		R_{ms} (A_{ms})	β_{Rcs}	B_{Ccs}	Mean failure rate (per yr)
S_1	Offsite power	0.20g	0.226	0.226	–
S_2	Condensate storage tank	0.24g	0.273	0.273	–
S_3	Reactor internals	0.67g	0.300	0.300	–
S_4	Reactor enclosure structure	1.05g	0.282	0.282	–
S_6	Reactor pressure vessel	1.25g	0.252	0.252	–
S_{10}	Standby liquid control system tank	1.33g	0.233	0.233	–
S_{11}	440-V bus/steam generator breakers	1.46g	0.411	0.411	–
S_{12}	440-V bus transformer breaker	1.49g	0.397	0.397	–
S_{13}	125/250-V DC bus	1.49g	0.397	0.397	–
S_{14}	4-kV bus/steam generator	1.49g	0.397	0.397	–
S_{15}	Diesel generator circuit	1.56g	0.368	0.368	–
S_{16}	Diesel generator heat and vent	1.55g	0.363	0.363	–
S_{17}	Residual heat removal system heat exchangers	1.09g	0.330	0.330	–
DG_R	DGR – diesel generator common mode	–	–	–	0.00125
W_R	WR – containment heat removal	–	–	–	0.00026
C_R	CR – scram system mechanical failure	–	–	–	1.00E-05
SLC_R	SLCR – standby liquid control	–	–	–	0.01

Table 2

Tsunami fragility of LGS NPP components (adapted from Ref. [19]).

Component		R_{mt} (A_{mt})	β_{Rct}	B_{Cct}
S_1	Offsite power	10 m	0.354	0.354
S_2	Condensate storage tank	10 m	0.212	0.212
S_{11}	440-V bus/SG breakers	11 m	0.212	0.212
S_{12}	440-V bus transformer breaker	11 m	0.212	0.212
S_{13}	125/250-V DC bus	11 m	0.212	0.212
S_{14}	4-kV bus/SG	11 m	0.212	0.212
S_{15}	Diesel generator circuit	11 m	0.212	0.212
S_{17}	RHR heat exchangers	10 m	0.212	0.212

Kwag, S., Ha, J. G., Kim, M. K., & Kim, J. H. (2019). Development of efficient external multi-hazard risk quantification methodology for nuclear facilities. Energies, 12(20), 3925.
Choi, E., Kwag, S., Ha, J. G., & Hahm, D. (2021). Development of a two-stage DQFM to improve efficiency of single-and multi-hazard risk quantification for nuclear facilities. Energies, 14(4), 1017.
Choi, E., Kwag, S., & Hahm, D. (2024). Multihazard capacity optimization of an NPP using a multi-objective genetic algorithm and sampling-based PSA. Nuclear Engineering and Technology, 56(2), 644-654.
Choi, E., Kwag, S., Kim, J. H., Ha, J. G., Hahm, D., & Kim, M. (2024). A review of COHRISK: Multihazard risk quantification software for nuclear power plants. Nuclear Engineering and Technology, 56(12), 5281-5290.

$$\begin{cases} T_sE_sUX = \left(S_{1s} \cup S_{1t}\right) \cap A; \\ A = \left(S_{11s} \cup S_{11t}\right) \cup \left(S_{12s} \cup S_{12t}\right) \cup \left(S_{13s} \cup S_{13t}\right) \cup \left(S_{14s} \cup S_{14t}\right) \cup \left(S_{15s} \cup S_{15t}\right) \cup \left(S_{16s} \cup S_{16t}\right) \cup DG_R \\ T_sR_b = \left(S_{4s} \cup S_{4t}\right) \\ T_sR_{pv} = \left(S_{6s} \cup S_{6t}\right) \\ T_sE_sC_mC_2 = \left(S_{1s} \cup S_{1t}\right) \cap \left(\left(S_{3s} \cup S_{3t}\right) \cup C_R\right) \cap \left(A \cup \left(S_{10s} \cup S_{10t}\right) \cup SLC_R\right) \\ T_sR_bC_m = \left(S_{4s} \cup S_{4t}\right) \cap \left(C_R \cup \left(S_{3s} \cup S_{3t}\right)\right) \\ T_sE_bW_m = \left(S_{1s} \cup S_{1t}\right) \cap \overline{A} \cap \left[\overline{\left(S_{17s} \cup S_{17t}\right)} \cap W_R \cup \overline{\left(S_{2s} \cup S_{2t}\right)} \cap \left(S_{17s} \cup S_{17t}\right)\right] \end{cases}$$

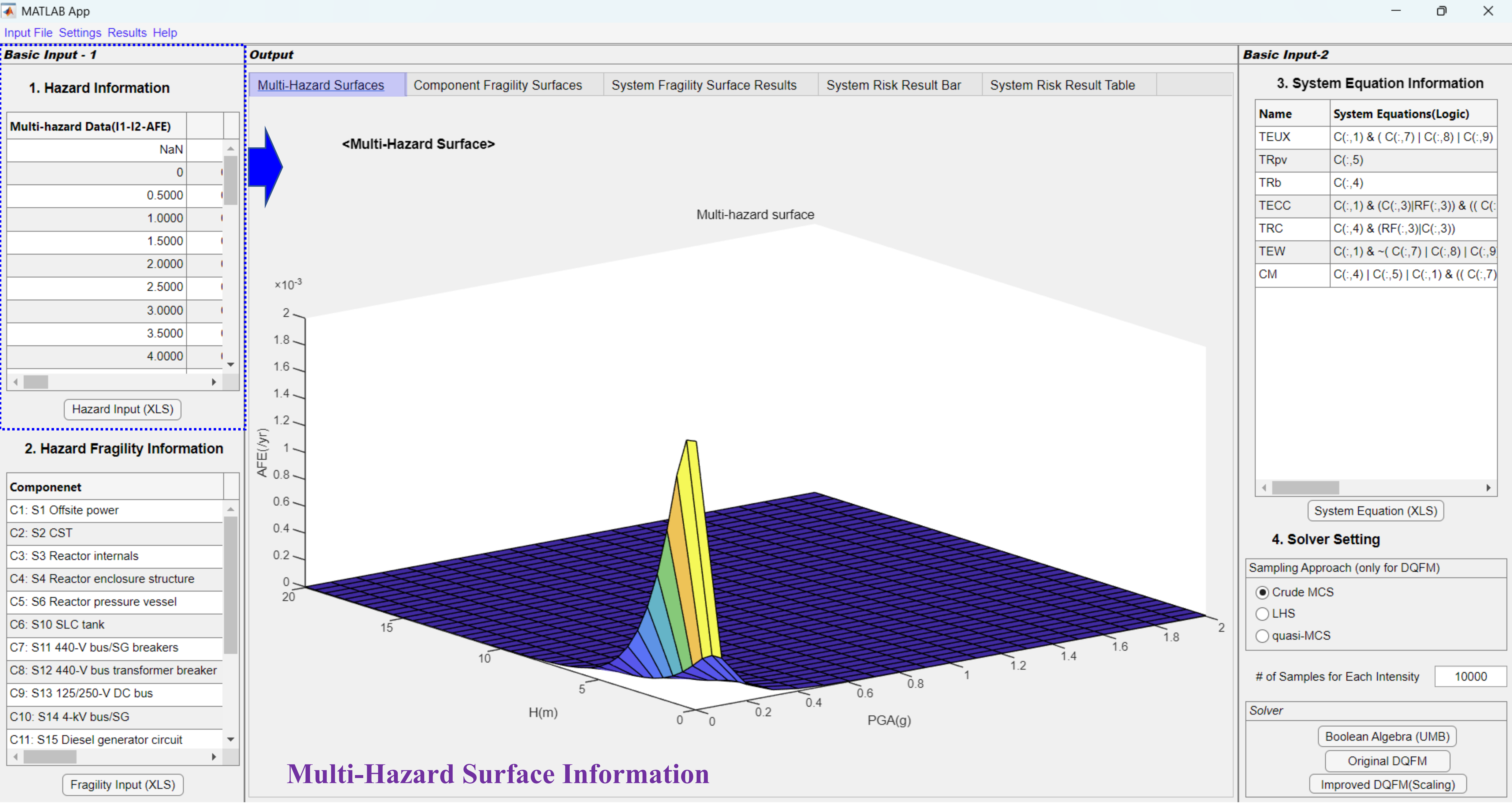
Core Melt Damage Sequence:

$$\begin{aligned} CM = {} & \left(S_{4s} \cup S_{4t}\right) \cup \left(S_{6s} \cup S_{6t}\right) \cup \left(S_{1s} \cup S_{1t}\right) \\ & \cap \left\{A \cup \left[\left(S_{3s} \cup S_{3t}\right) \cup C_R\right] \cap \left[\left(S_{10s} \cup S_{10t}\right) \cup SLC_R\right] \cup \left[\left(S_{17s} \cup S_{17t}\right) \cup W_R\right]\right\} \end{aligned}$$

Kwag, S., Ha, J. G., Kim, M. K., & Kim, J. H. (2019). Development of efficient external multi-hazard risk quantification methodology for nuclear facilities. Energies, 12(20), 3925.
Choi, E., Kwag, S., Ha, J. G., & Hahm, D. (2021). Development of a two-stage DQFM to improve efficiency of single-and multi-hazard risk quantification for nuclear facilities. Energies, 14(4), 1017.
Choi, E., Kwag, S., & Hahm, D. (2024). Multihazard capacity optimization of an NPP using a multi-objective genetic algorithm and sampling-based PSA. Nuclear Engineering and Technology, 56(2), 644-654.
Choi, E., Kwag, S., Kim, J. H., Ha, J. G., Hahm, D., & Kim, M. (2024). A review of COHRISK: Multihazard risk quantification software for nuclear power plants. Nuclear Engineering and Technology, 56(12), 5281-5290.

Example: LGS Nuclear Power Plant

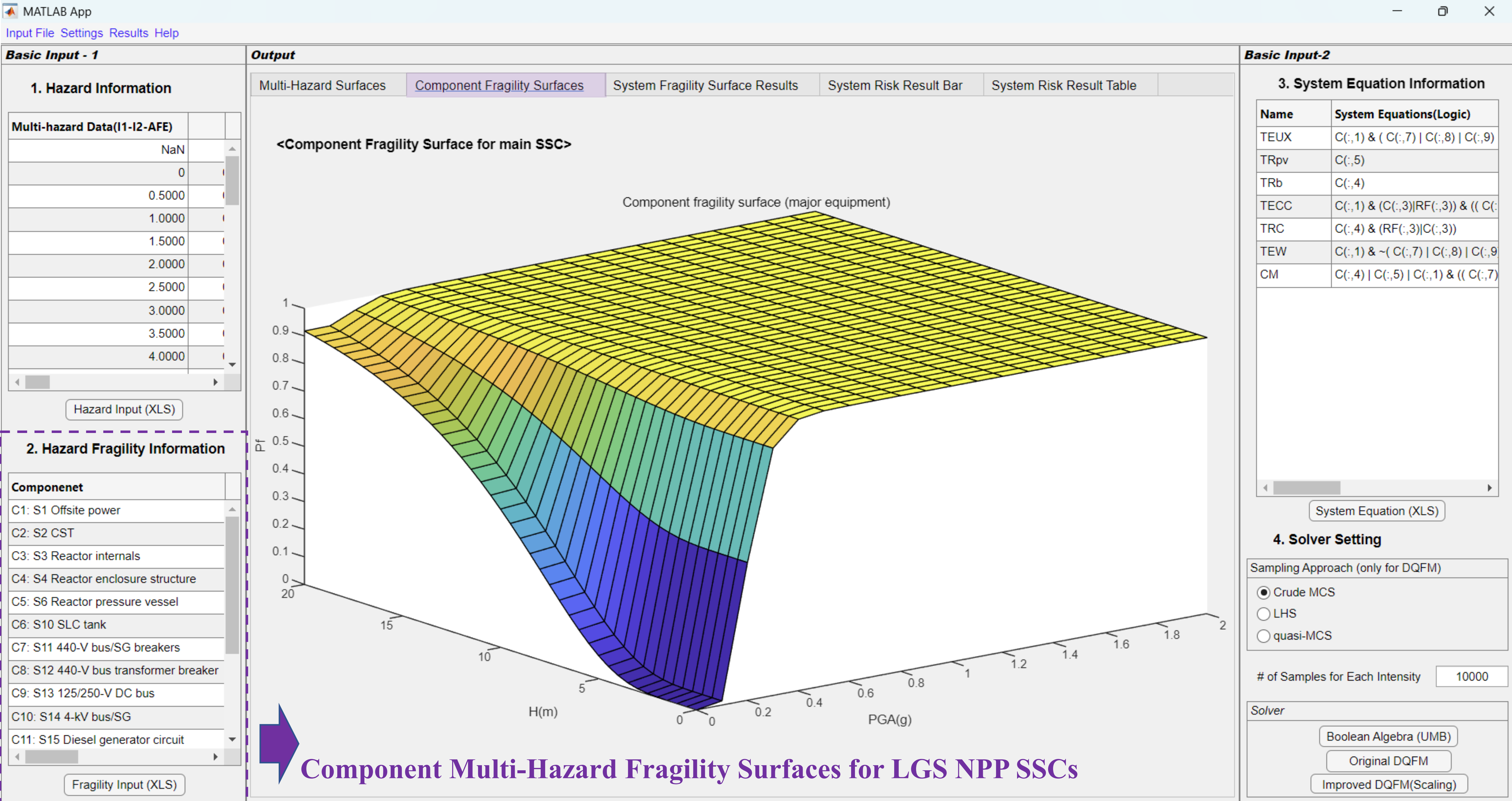
Multi-Harzard Surface Input & Confirmation



Multi-Hazard Surface Information

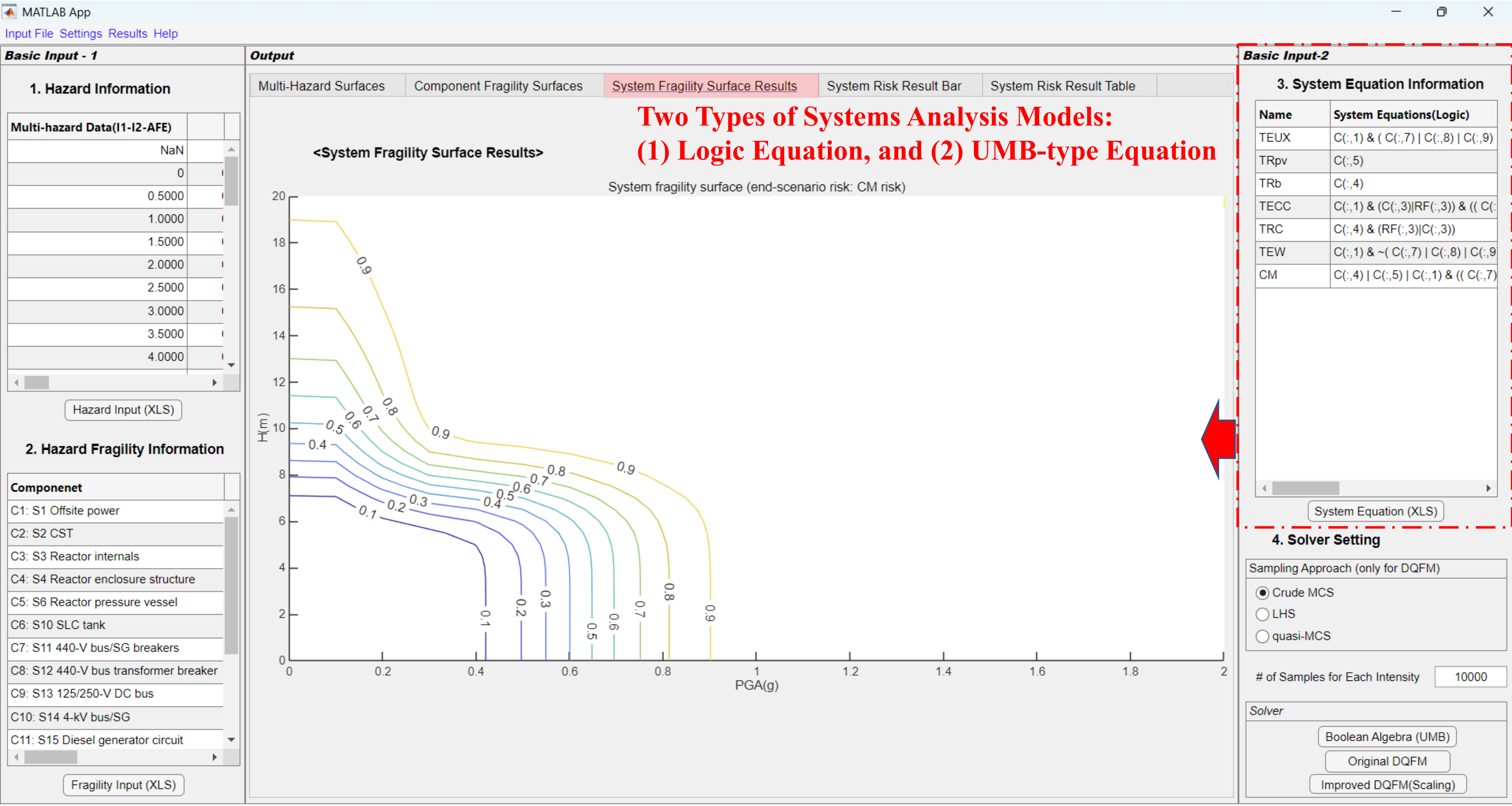
Example: LGS Nuclear Power Plant

Component Multi-Hazard Fragility Surface Input, Calculation & Confirmation



Component Multi-Hazard Fragility Surfaces for LGS NPP SSCs

Example: LGS Nuclear Power Plant Systems Analysis Model Input & System Multi-Hazard Fragility Surface Results



Example: LGS Nuclear Power Plant

Risk Quantification Bar Result using Three Different Solvers and Three Different Samplers

Example: LGS Nuclear Power Plant

MATLAB App

Input File Settings Results Help

Basic Input - 1

1. Hazard Information

Multi-hazard Data(I1-I2-AFE)
NaN
0
0.5000
1.0000
1.5000
2.0000
2.5000
3.0000
3.5000
4.0000

Hazard Input (XLS)

2. Hazard Fragility Information

Componenet
C1: S1 Offsite power
C2: S2 CST
C3: S3 Reactor internals
C4: S4 Reactor enclosure structure
C5: S6 Reactor pressure vessel
C6: S10 SLC tank
C7: S11 440-V bus/SG breakers
C8: S12 440-V bus transformer breaker
C9: S13 125/250-V DC bus
C10: S14 4-kV bus/SG
C11: S15 Diesel generator circuit

Fragility Input (XLS)

Output

Multi-Hazard Surfaces | Component Fragility Surfaces | System Fragility Surface Results | System Risk Result Bar | System Risk Result Table

<System Risk Result Table>

Name	Risk(/yr)	CDF contribution(/yr)
TEUX	7.3966e-06	7.3966e-06
TRpv	4.0996e-07	4.0996e-07
TRb	1.0397e-06	1.0397e-06
TECC	1.4091e-06	1.4091e-06
TRC	5.8330e-07	5.8330e-07
TEW	7.1350e-07	7.1350e-07
CM	9.6242e-06	9.6242e-06

Risk Quantification Bar Result using Three Different Solvers and Three Different Samplers

Basic Input-2

3. System Equation Information

Name	System Equations(Logic)
TEUX	C(:,1) & (C(:,7) | C(:,8) | C(:,9)
TRpv	C(:,5)
TRb	C(:,4)
TECC	C(:,1) & (C(:,3)|RF(:,3)) & ((C(:
TRC	C(:,4) & (RF(:,3)|C(:,3))
TEW	C(:,1) & ~(C(:,7) | C(:,8) | C(:,9
CM	C(:,4) | C(:,5) | C(:,1) & ((C(:,7)

System Equation (XLS)

4. Solver Setting

Sampling Approach (only for DQFM)

- (•) Crude MCS
- ( ) LHS
- ( ) quasi-MCS

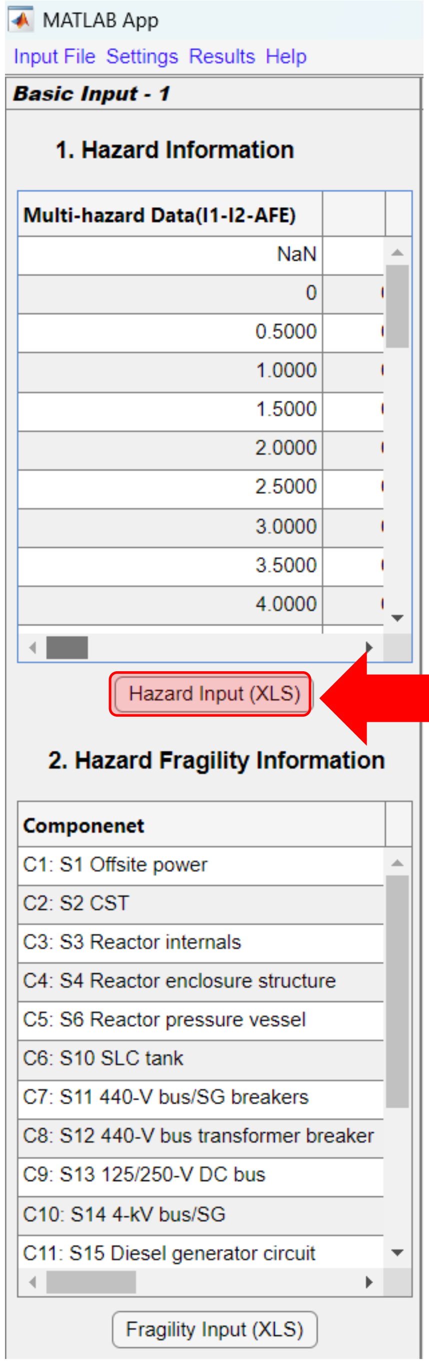
\# of Samples for Each Intensity: 10000

Solver

- Boolean Algebra (UMB)
- Original DQFM
- Improved DQFM(Scaling)

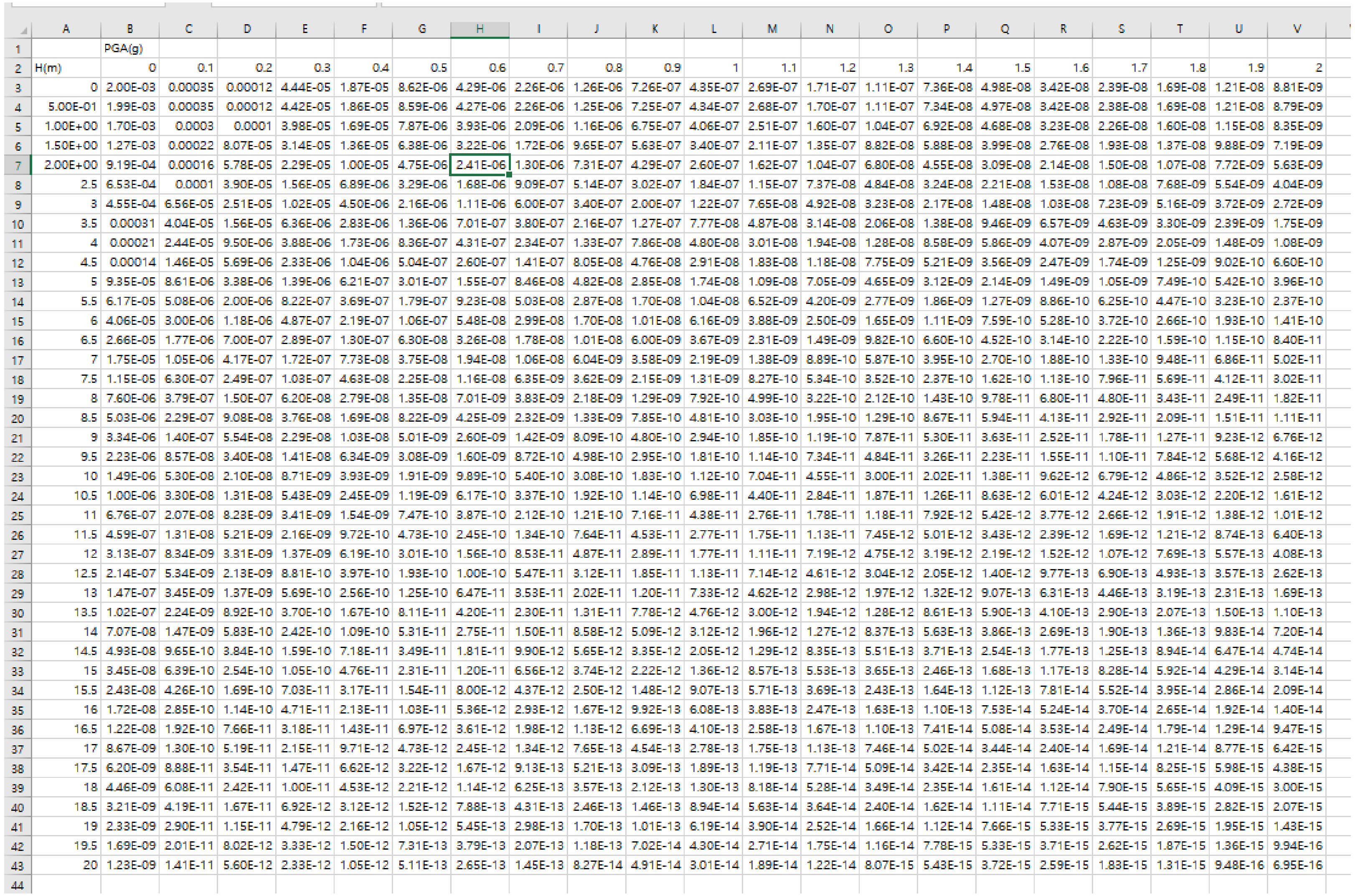
Appendix. File Type and Import for Example Case

Appendix. File Type and Import



Input_Data.xlsx

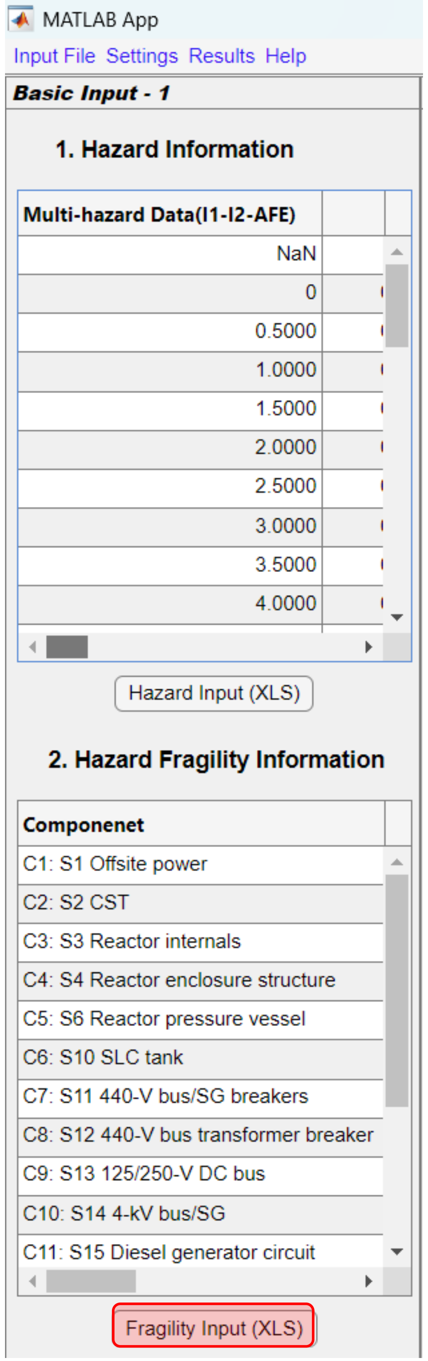
Button Click and Import Sheet1 of the EXCEL file.



	A	B	C	D	E	F	G	H	I	J	K
1		PGA(g)									
2	H(m)	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
3	0	2.00E-03	0.00035	0.00012	4.44E-05	1.87E-05	8.62E-06	4.29E-06	2.26E-06	1.26E-06	7.26E-07
4	5.00E-01	1.99E-03	0.00035	0.00012	4.42E-05	1.86E-05	8.59E-06	4.27E-06	2.26E-06	1.25E-06	7.25E-07
5	1.00E+00	1.70E-03	0.0003	0.0001	3.98E-05	1.69E-05	7.87E-06	3.93E-06	2.09E-06	1.16E-06	6.75E-07
6	1.50E+00	1.27E-03	0.00022	8.07E-05	3.14E-05	1.36E-05	6.38E-06	3.22E-06	1.72E-06	9.65E-07	5.63E-07
7	2.00E+00	9.19E-04	0.00016	5.78E-05	2.29E-05	1.00E-05	4.75E-06	2.41E-06	1.30E-06	7.31E-07	4.29E-07
8	2.5	6.53E-04	0.0001	3.90E-05	1.56E-05	6.89E-06	3.29E-06	1.68E-06	9.09E-07	5.14E-07	3.02E-07
9	3	4.55E-04	6.56E-05	2.51E-05	1.02E-05	4.50E-06	2.16E-06	1.11E-06	6.00E-07	3.40E-07	2.00E-07
10	3.5	0.00031	4.04E-05	1.56E-05	6.36E-06	2.83E-06	1.36E-06	7.01E-07	3.80E-07	2.16E-07	1.27E-07
11	4	0.00021	2.44E-05	9.50E-06	3.88E-06	1.73E-06	8.36E-07	4.31E-07	2.34E-07	1.33E-07	7.86E-08
12	4.5	0.00014	1.46E-05	5.69E-06	2.33E-06	1.04E-06	5.04E-07	2.60E-07	1.41E-07	8.05E-08	4.76E-08
13	5	9.35E-05	8.61E-06	3.38E-06	1.39E-06	6.21E-07	3.01E-07	1.55E-07	8.46E-08	4.82E-08	2.85E-08
14	5.5	6.17E-05	5.08E-06	2.00E-06	8.22E-07	3.69E-07	1.79E-07	9.23E-08	5.03E-08	2.87E-08	1.70E-08
15	6	4.06E-05	3.00E-06	1.18E-06	4.87E-07	2.19E-07	1.06E-07	5.48E-08	2.99E-08	1.70E-08	1.01E-08
16	6.5	2.66E-05	1.77E-06	7.00E-07	2.89E-07	1.30E-07	6.30E-08	3.26E-08	1.78E-08	1.01E-08	6.00E-09
17	7	1.75E-05	1.05E-06	4.17E-07	1.72E-07	7.73E-08	3.75E-08	1.94E-08	1.06E-08	6.04E-09	3.58E-09
18	7.5	1.15E-05	6.30E-07	2.49E-07	1.03E-07	4.63E-08	2.25E-08	1.16E-08	6.35E-09	3.62E-09	2.15E-09
19	8	7.60E-06	3.79E-07	1.50E-07	6.20E-08	2.79E-08	1.35E-08	7.01E-09	3.83E-09	2.18E-09	1.29E-09
20	8.5	5.03E-06	2.29E-07	9.08E-08	3.76E-08	1.69E-08	8.22E-09	4.25E-09	2.32E-09	1.33E-09	7.85E-10
21	9	3.34E-06	1.40E-07	5.54E-08	2.29E-08	1.03E-08	5.01E-09	2.60E-09	1.42E-09	8.09E-10	4.80E-10
22	9.5	2.23E-06	8.57E-08	3.40E-08	1.41E-08	6.34E-09	3.08E-09	1.60E-09	8.72E-10	4.98E-10	2.95E-10
23	10	1.49E-06	5.30E-08	2.10E-08	8.71E-09	3.93E-09	1.91E-09	9.89E-10	5.40E-10	3.08E-10	1.83E-10
24	10.5	1.00E-06	3.30E-08	1.31E-08	5.43E-09	2.45E-09	1.19E-09	6.17E-10	3.37E-10	1.92E-10	1.14E-10
25	11	6.76E-07	2.07E-08	8.23E-09	3.41E-09	1.54E-09	7.47E-10	3.87E-10	2.12E-10	1.21E-10	7.16E-11
26	11.5	4.59E-07	1.31E-08	5.21E-09	2.16E-09	9.72E-10	4.73E-10	2.45E-10	1.34E-10	7.64E-11	4.53E-11
27	12	3.13E-07	8.34E-09	3.31E-09	1.37E-09	6.19E-10	3.01E-10	1.56E-10	8.53E-11	4.87E-11	2.89E-11
28	12.5	2.14E-07	5.34E-09	2.13E-09	8.81E-10	3.97E-10	1.93E-10	1.00E-10	5.47E-11	3.12E-11	1.85E-11
29	13	1.47E-07	3.45E-09	1.37E-09	5.69E-10	2.56E-10	1.25E-10	6.47E-11	3.53E-11	2.02E-11	1.20E-11
30	13.5	1.02E-07	2.24E-09	8.92E-10	3.70E-10	1.67E-10	8.11E-11	4.20E-11	2.30E-11	1.31E-11	7.78E-12
31	14	7.07E-08	1.47E-09	5.83E-10	2.42E-10	1.09E-10	5.31E-11	2.75E-11	1.50E-11	8.58E-12	5.09E-12
32	14.5	4.93E-08	9.65E-10	3.84E-10	1.59E-10	7.18E-11	3.49E-11	1.81E-11	9.90E-12	5.65E-12	3.35E-12
33	15	3.45E-08	6.39E-10	2.54E-10	1.05E-10	4.76E-11	2.31E-11	1.20E-11	6.56E-12	3.74E-12	2.22E-12
34	15.5	2.43E-08	4.26E-10	1.69E-10	7.03E-11	3.17E-11	1.54E-11	8.00E-12	4.37E-12	2.50E-12	1.48E-12
35	16	1.72E-08	2.85E-10	1.14E-10	4.71E-11	2.13E-11	1.03E-11	5.36E-12	2.93E-12	1.67E-12	9.92E-13
36	16.5	1.22E-08	1.92E-10	7.66E-11	3.18E-11	1.43E-11	6.97E-12	3.61E-12	1.98E-12	1.13E-12	6.69E-13
37	17	8.67E-09	1.30E-10	5.19E-11	2.15E-11	9.71E-12	4.73E-12	2.45E-12	1.34E-12	7.65E-13	4.54E-13
38	17.5	6.20E-09	8.88E-11	3.54E-11	1.47E-11	6.62E-12	3.22E-12	1.67E-12	9.13E-13	5.21E-13	3.09E-13
39	18	4.46E-09	6.08E-11	2.42E-11	1.00E-11	4.53E-12	2.21E-12	1.14E-12	6.25E-13	3.57E-13	2.12E-13
40	18.5	3.21E-09	4.19E-11	1.67E-11	6.92E-12	3.12E-12	1.52E-12	7.88E-13	4.31E-13	2.46E-13	1.46E-13
41	19	2.33E-09	2.90E-11	1.15E-11	4.79E-12	2.16E-12	1.05E-12	5.45E-13	2.98E-13	1.70E-13	1.01E-13
42	19.5	1.69E-09	2.01E-11	8.02E-12	3.33E-12	1.50E-12	7.31E-13	3.79E-13	2.07E-13	1.18E-13	7.02E-14
43	20	1.23E-09	1.41E-11	5.60E-12	2.33E-12	1.05E-12	5.11E-13	2.65E-13	1.45E-13	8.27E-14	4.91E-14
44											

	A	L	M	N	O	P	Q	R	S	T	U	V
1												
2	H(m)	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
3	0	4.35E-07	2.69E-07	1.71E-07	1.11E-07	7.36E-08	4.98E-08	3.42E-08	2.39E-08	1.69E-08	1.21E-08	8.81E-09
4	5.00E-01	4.34E-07	2.68E-07	1.70E-07	1.11E-07	7.34E-08	4.97E-08	3.42E-08	2.38E-08	1.69E-08	1.21E-08	8.79E-09
5	1.00E+00	4.06E-07	2.51E-07	1.60E-07	1.04E-07	6.92E-08	4.68E-08	3.23E-08	2.26E-08	1.60E-08	1.15E-08	8.35E-09
6	1.50E+00	3.40E-07	2.11E-07	1.35E-07	8.82E-08	5.88E-08	3.99E-08	2.76E-08	1.93E-08	1.37E-08	9.88E-09	7.19E-09
7	2.00E+00	2.60E-07	1.62E-07	1.04E-07	6.80E-08	4.55E-08	3.09E-08	2.14E-08	1.50E-08	1.07E-08	7.72E-09	5.63E-09
8	2.5	1.84E-07	1.15E-07	7.37E-08	4.84E-08	3.24E-08	2.21E-08	1.53E-08	1.08E-08	7.68E-09	5.54E-09	4.04E-09
9	3	1.22E-07	7.65E-08	4.92E-08	3.23E-08	2.17E-08	1.48E-08	1.03E-08	7.23E-09	5.16E-09	3.72E-09	2.72E-09
10	3.5	7.77E-08	4.87E-08	3.14E-08	2.06E-08	1.38E-08	9.46E-09	6.57E-09	4.63E-09	3.30E-09	2.39E-09	1.75E-09
11	4	4.80E-08	3.01E-08	1.94E-08	1.28E-08	8.58E-09	5.86E-09	4.07E-09	2.87E-09	2.05E-09	1.48E-09	1.08E-09
12	4.5	2.91E-08	1.83E-08	1.18E-08	7.75E-09	5.21E-09	3.56E-09	2.47E-09	1.74E-09	1.25E-09	9.02E-10	6.60E-10
13	5	1.74E-08	1.09E-08	7.05E-09	4.65E-09	3.12E-09	2.14E-09	1.49E-09	1.05E-09	7.49E-10	5.42E-10	3.96E-10
14	5.5	1.04E-08	6.52E-09	4.20E-09	2.77E-09	1.86E-09	1.27E-09	8.86E-10	6.25E-10	4.47E-10	3.23E-10	2.37E-10
15	6	6.16E-09	3.88E-09	2.50E-09	1.65E-09	1.11E-09	7.59E-10	5.28E-10	3.72E-10	2.66E-10	1.93E-10	1.41E-10
16	6.5	3.67E-09	2.31E-09	1.49E-09	9.82E-10	6.60E-10	4.52E-10	3.14E-10	2.22E-10	1.59E-10	1.15E-10	8.40E-11
17	7	2.19E-09	1.38E-09	8.89E-10	5.87E-10	3.95E-10	2.70E-10	1.88E-10	1.33E-10	9.48E-11	6.86E-11	5.02E-11
18	7.5	1.31E-09	8.27E-10	5.34E-10	3.52E-10	2.37E-10	1.62E-10	1.13E-10	7.96E-11	5.69E-11	4.12E-11	3.02E-11
19	8	7.92E-10	4.99E-10	3.22E-10	2.12E-10	1.43E-10	9.78E-11	6.80E-11	4.80E-11	3.43E-11	2.49E-11	1.82E-11
20	8.5	4.81E-10	3.03E-10	1.95E-10	1.29E-10	8.67E-11	5.94E-11	4.13E-11	2.92E-11	2.09E-11	1.51E-11	1.11E-11
21	9	2.94E-10	1.85E-10	1.19E-10	7.87E-11	5.30E-11	3.63E-11	2.52E-11	1.78E-11	1.27E-11	9.23E-12	6.76E-12
22	9.5	1.81E-10	1.14E-10	7.34E-11	4.84E-11	3.26E-11	2.23E-11	1.55E-11	1.10E-11	7.84E-12	5.68E-12	4.16E-12
23	10	1.12E-10	7.04E-11	4.55E-11	3.00E-11	2.02E-11	1.38E-11	9.62E-12	6.79E-12	4.86E-12	3.52E-12	2.58E-12
24	10.5	6.98E-11	4.40E-11	2.84E-11	1.87E-11	1.26E-11	8.63E-12	6.01E-12	4.24E-12	3.03E-12	2.20E-12	1.61E-12
25	11	4.38E-11	2.76E-11	1.78E-11	1.18E-11	7.92E-12	5.42E-12	3.77E-12	2.66E-12	1.91E-12	1.38E-12	1.01E-12
26	11.5	2.77E-11	1.75E-11	1.13E-11	7.45E-12	5.01E-12	3.43E-12	2.39E-12	1.69E-12	1.21E-12	8.74E-13	6.40E-13
27	12	1.77E-11	1.11E-11	7.19E-12	4.75E-12	3.19E-12	2.19E-12	1.52E-12	1.07E-12	7.69E-13	5.57E-13	4.08E-13
28	12.5	1.13E-11	7.14E-12	4.61E-12	3.04E-12	2.05E-12	1.40E-12	9.77E-13	6.90E-13	4.93E-13	3.57E-13	2.62E-13
29	13	7.33E-12	4.62E-12	2.98E-12	1.97E-12	1.32E-12	9.07E-13	6.31E-13	4.46E-13	3.19E-13	2.31E-13	1.69E-13
30	13.5	4.76E-12	3.00E-12	1.94E-12	1.28E-12	8.61E-13	5.90E-13	4.10E-13	2.90E-13	2.07E-13	1.50E-13	1.10E-13
31	14	3.12E-12	1.96E-12	1.27E-12	8.37E-13	5.63E-13	3.86E-13	2.69E-13	1.90E-13	1.36E-13	9.83E-14	7.20E-14
32	14.5	2.05E-12	1.29E-12	8.35E-13	5.51E-13	3.71E-13	2.54E-13	1.77E-13	1.25E-13	8.94E-14	6.47E-14	4.74E-14
33	15	1.36E-12	8.57E-13	5.53E-13	3.65E-13	2.46E-13	1.68E-13	1.17E-13	8.28E-14	5.92E-14	4.29E-14	3.14E-14
34	15.5	9.07E-13	5.71E-13	3.69E-13	2.43E-13	1.64E-13	1.12E-13	7.81E-14	5.52E-14	3.95E-14	2.86E-14	2.09E-14
35	16	6.08E-13	3.83E-13	2.47E-13	1.63E-13	1.10E-13	7.53E-14	5.24E-14	3.70E-14	2.65E-14	1.92E-14	1.40E-14
36	16.5	4.10E-13	2.58E-13	1.67E-13	1.10E-13	7.41E-14	5.08E-14	3.53E-14	2.49E-14	1.79E-14	1.29E-14	9.47E-15
37	17	2.78E-13	1.75E-13	1.13E-13	7.46E-14	5.02E-14	3.44E-14	2.40E-14	1.69E-14	1.21E-14	8.77E-15	6.42E-15
38	17.5	1.89E-13	1.19E-13	7.71E-14	5.09E-14	3.42E-14	2.35E-14	1.63E-14	1.15E-14	8.25E-15	5.98E-15	4.38E-15
39	18	1.30E-13	8.18E-14	5.28E-14	3.49E-14	2.35E-14	1.61E-14	1.12E-14	7.90E-15	5.65E-15	4.09E-15	3.00E-15
40	18.5	8.94E-14	5.63E-14	3.64E-14	2.40E-14	1.62E-14	1.11E-14	7.71E-15	5.44E-15	3.89E-15	2.82E-15	2.07E-15
41	19	6.19E-14	3.90E-14	2.52E-14	1.66E-14	1.12E-14	7.66E-15	5.33E-15	3.77E-15	2.69E-15	1.95E-15	1.43E-15
42	19.5	4.30E-14	2.71E-14	1.75E-14	1.16E-14	7.78E-15	5.33E-15	3.71E-15	2.62E-15	1.87E-15	1.36E-15	9.94E-16
43	20	3.01E-14	1.89E-14	1.22E-14	8.07E-15	5.43E-15	3.72E-15	2.59E-15	1.83E-15	1.31E-15	9.48E-16	6.95E-16
44												

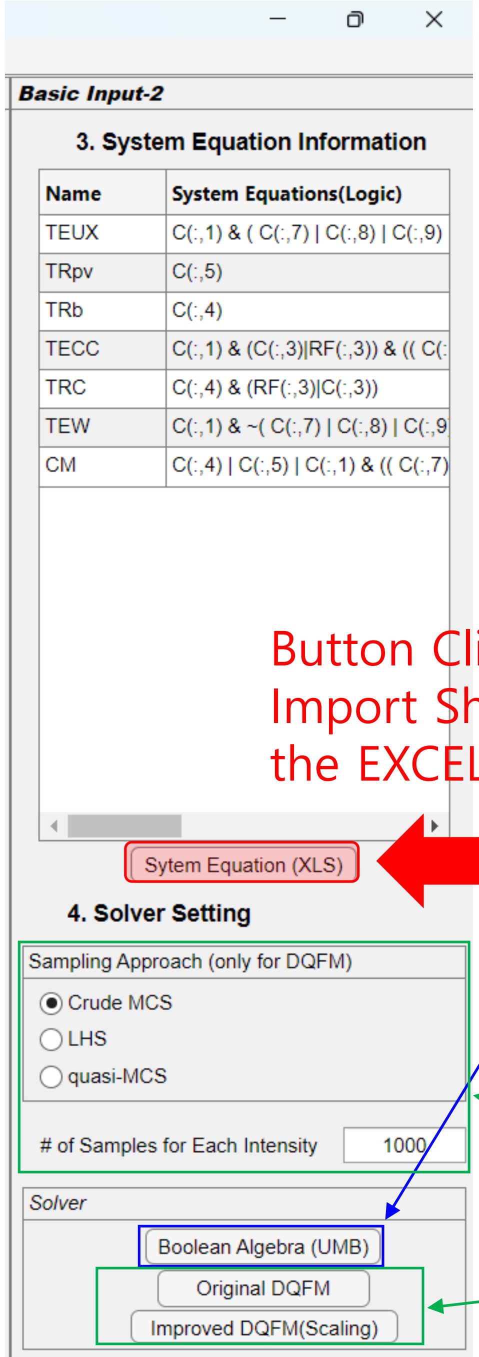
Appendix. File Type and Import



Input_Data.xlsx

	A	B	C	D	E	F	G	H
1		PGA			H			
2		Am	br	bu	Hm	br	bu	random_failure
3	C1: S1 Offsite power	0.2	0.2	0.25	10	0.5	0	0
4	C2: S2 CST	0.24	0.23	0.31	10	0.3	0	0
5	C3: S3 Reactor internals	0.67	0.28	0.32	1E+10	0	0	0
6	C4: S4 Reactor enclosure structure	1.05	0.31	0.25	1E+10	0	0	0
7	C5: S6 Reactor pressure vessel	1.25	0.28	0.22	1E+10	0	0	0
8	C6: S10 SLC tank	1.33	0.27	0.19	1E+10	0	0	0
9	C7: S11 440-V bus/SG breakers	1.46	0.38	0.44	11	0.3	0	0
10	C8: S12 440-V bus transformer breaker	1.49	0.36	0.43	11	0.3	0	0
11	C9: S13 125/250-V DC bus	1.49	0.36	0.43	11	0.3	0	0
12	C10: S14 4-kV bus/SG	1.49	0.36	0.43	11	0.3	0	0
13	C11: S15 Diesel generator circuit	1.56	0.32	0.41	11	0.3	0	0
14	C12: S16 Diesel generator heat and vent	1.55	0.28	0.43	1E+10	0	0	0
15	C13: S17 RHR heat exchangers	1.09	0.32	0.34	10	0.3	0	0
16	RF1: DGR Diesel generator common mod	0	0	0	0	0	0	0.00125
17	RF2: WR Containment heat removal	0	0	0	0	0	0	0.00026
18	RF3: CR Scram system mechanical failure	0	0	0	0	0	0	1.00E-05
19	RF4: SLCR Standby liquid control	0	0	0	0	0	0	0.01

Button Click and Import Sheet2 of the EXCEL file.



Input_Data.xlsx

<table>
<tr><th></th><th>A</th><th>B</th></tr>
<tr><td>1</td><td>Name</td><td>System Equations (Logic Tree)</td></tr>
<tr><td>2</td><td>TEUX</td><td>C(:,1) & (C(:,7) | C(:,8) | C(:,9) | C(:,10) | C(:,11) | C(:,12) | RF(:,1))</td></tr>
<tr><td>3</td><td>TRpv</td><td>C(:,5)</td></tr>
<tr><td>4</td><td>TRb</td><td>C(:,4)</td></tr>
<tr><td>5</td><td>TECC</td><td>C(:,1) & (C(:,3)|RF(:,3)) & ((C(:,7) | C(:,8) | C(:,9) | C(:,10) | C(:,11) | C(:,12) | RF(:,1))|C(:,6)|RF(:,4))</td></tr>
<tr><td>6</td><td>TRC</td><td>C(:,4) & (RF(:,3)|C(:,3))</td></tr>
<tr><td>7</td><td>TEW</td><td>C(:,1) & ~(C(:,7) | C(:,8) | C(:,9) | C(:,10) | C(:,11) | C(:,12) | RF(:,1)) & ((~C(:,13) & RF(:,2)) | (~C(:,2) & C(:,13)))</td></tr>
<tr><td>8</td><td>CM</td><td>C(:,4) | C(:,5) | C(:,1) & ((C(:,7) | C(:,8) | C(:,9) | C(:,10) | C(:,11) | C(:,12) | RF(:,1)) | (C(:,3)|RF(:,3))&(C(:,6)|RF(:,4)) | (C(:,13)|RF(:,2)))</td></tr>
</table>

“Event Tree and Fault Tree-type” Model -> “Logical Expression”

⋮

Button Click and Import Sheet3 of the EXCEL file.

	C	D
1	System Equations (UMB)	Secondary ET
2	C(:,:,1).*(1 - (1 -C(:,:,7)).*(1 -C(:,:,8)).*(1 -C(:,:,9)).*(1 -C(:,:,10)).*(1 -C(:,:,11)).*(1 -C(:,:,12)).*(1 -RF(:,:,1)))	1
3	C(:,:,5)	1
4	C(:,:,4)	1
5	C(:,:,1).* (1 -(1 -C(:,:,3)).*(1 -RF(:,:,3))).* (1 -(1 -(1 - (1 -C(:,:,7)).*(1 -C(:,:,8)).*(1 -C(:,:,9)).*(1 -C(:,:,10)).*(1 -C(:,:,11)).*(1 -C(:,:,12)).*(1 -RF(:,:,1)))).*(1 -C(:,:,6)).*(1 -RF(:,:,4)))	1
6	C(:,:,4).* (1 -(1 -RF(:,:,3)).*(1 -C(:,:,3)))	1
7	C(:,:,1).* (1 -(1 - (1 -C(:,:,7)).*(1 -C(:,:,8)).*(1 -C(:,:,9)).*(1 -C(:,:,10)).*(1 -C(:,:,11)).*(1 -C(:,:,12)).*(1 -RF(:,:,1)))) .*(1 -(1 -(1 -C(:,:,13)).*RF(:,:,2)).*(1 -(1 -C(:,:,2)).*C(:,:,13)))	1
8	1 -(1 -C(:,:,4)).*(1 -C(:,:,5)).*(1 -C(:,:,1).*(1 -(1 -(1 - (1 -C(:,:,7)).*(1 -C(:,:,8)).*(1 -C(:,:,9)).*(1 -C(:,:,10)).*(1 -C(:,:,11)).*(1 -C(:,:,12)).*(1 -RF(:,:,1)))).*(1 -((1 -(1 -C(:,:,3)).*(1 -RF(:,:,3))).*(1 -(1 -C(:,:,6)).*(1 -RF(:,:,4))))).*(1 -(1 -(1 -C(:,:,13)).*(1 -RF(:,:,2))))))	1

“Event Tree and Fault Tree-type” Model ->
“Uni-Modal Bound(UMB) Approach-type Mathematical Expression”